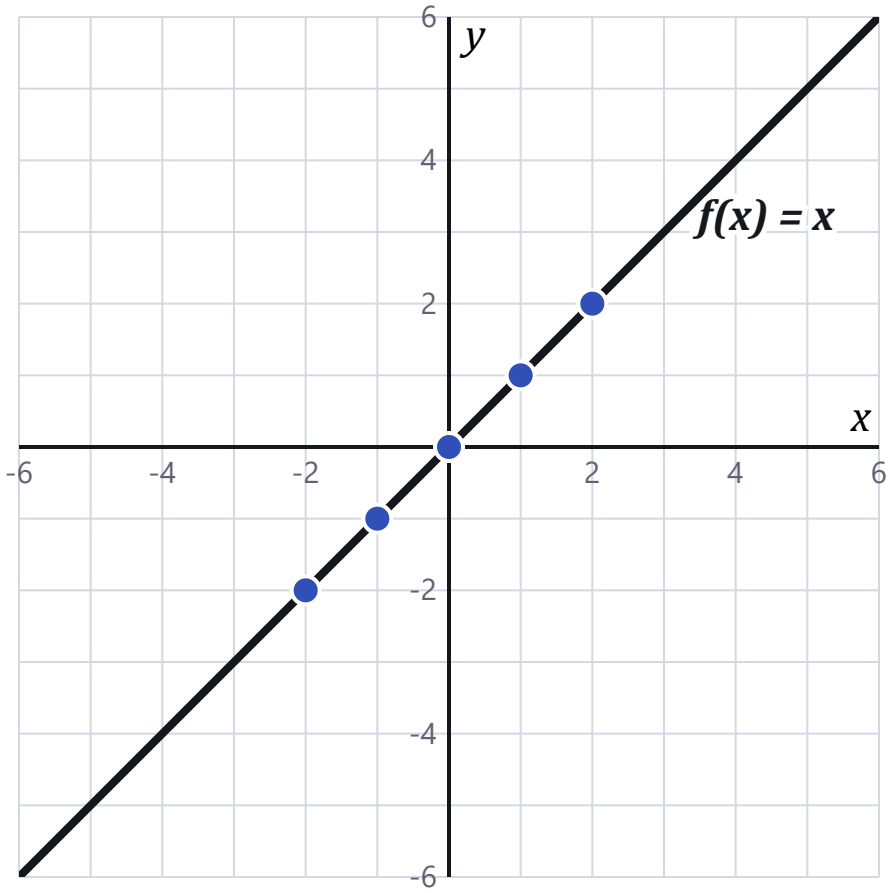
The Parent Function

$$f(x) = x$$



x	$f(x)$
-2	-2
-1	-1
0	0
1	1
2	2

Every output equals its input.

Slope

$m = 1$

Up 1, right 1

y-intercept

$(0, 0)$

Passes through the origin

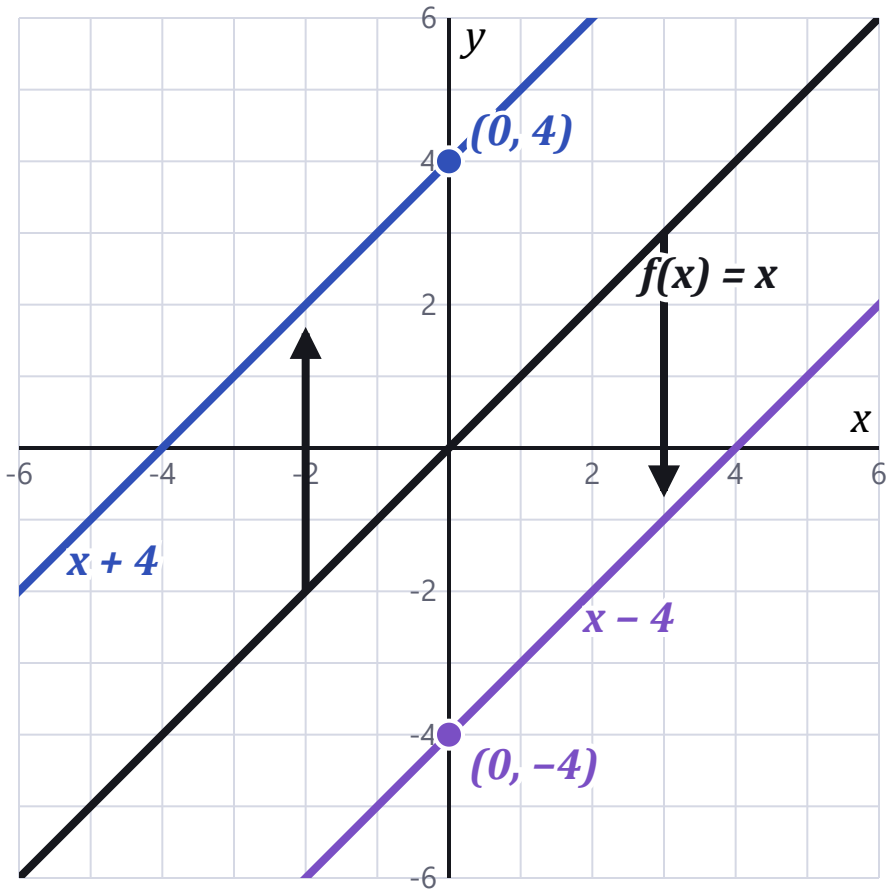
Domain, range

All reals

A **parent function** is the simplest function in its family. Every line is a transformation of it.

Vertical Translation

$$y = f(x) + d$$



$d > 0$

Moves **up**.
$f(x) + 4$: every point up 4.

$d < 0$

Moves **down**.
$f(x) - 4$: every point down 4.

What changes

The **outputs** (y-values) and the ***y*-intercept**, which moves to $(0, d)$.

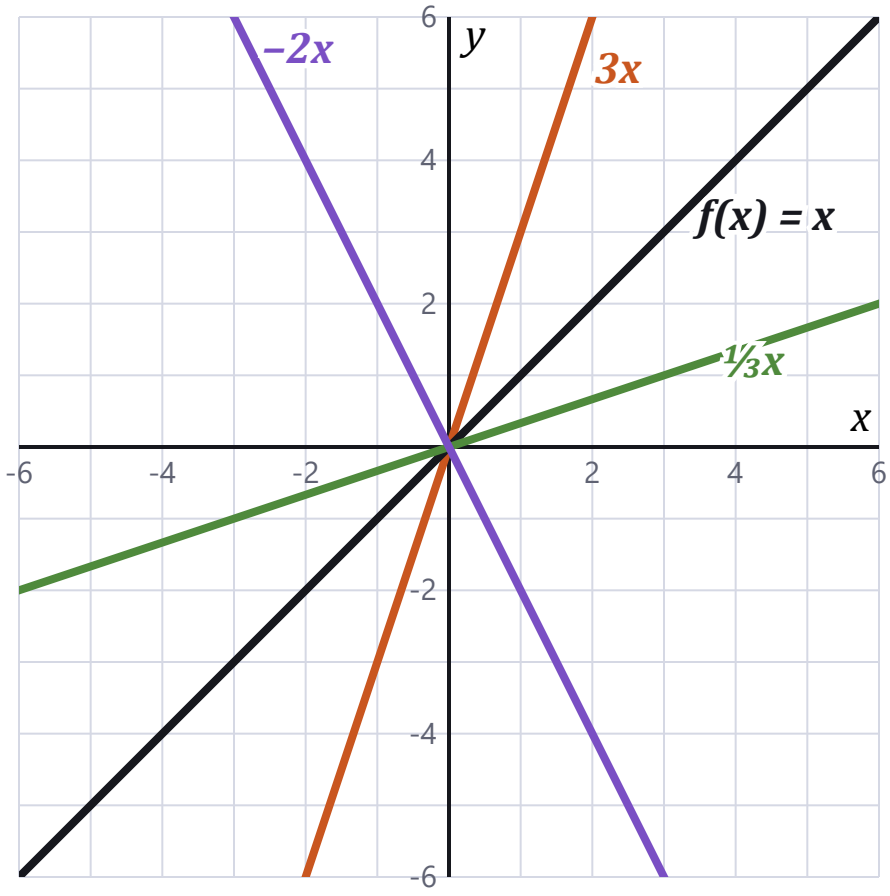
What stays

The **slope**. A line and its vertical translation are **parallel**: they never meet.

Example: $j(x) = 2x - 1$ up 4 → $j(x) + 4 = 2x + 3$

Vertical Dilation

$$y = a \cdot f(x)$$



$|a| > 1$

Stretch: steeper.
$3 \cdot f(x) = 3x$

$0 < |a| < 1$

Compress: flatter.
$\frac{1}{3} \cdot f(x) = \frac{1}{3}x$

$a < 0$

Reflect across the x-axis.
$-2 \cdot f(x) = -2x$

What changes

Every **output** is multiplied by a.
For $f(x) = x$, the slope becomes a.

What stays

The origin $(0, 0)$ doesn't move:
$a \cdot 0 = 0$.

> Example: $(2, 5)$ is on $f(x)$, so $(2, -10)$ is on $-2 \cdot f(x)$.

Graph Any Transformation

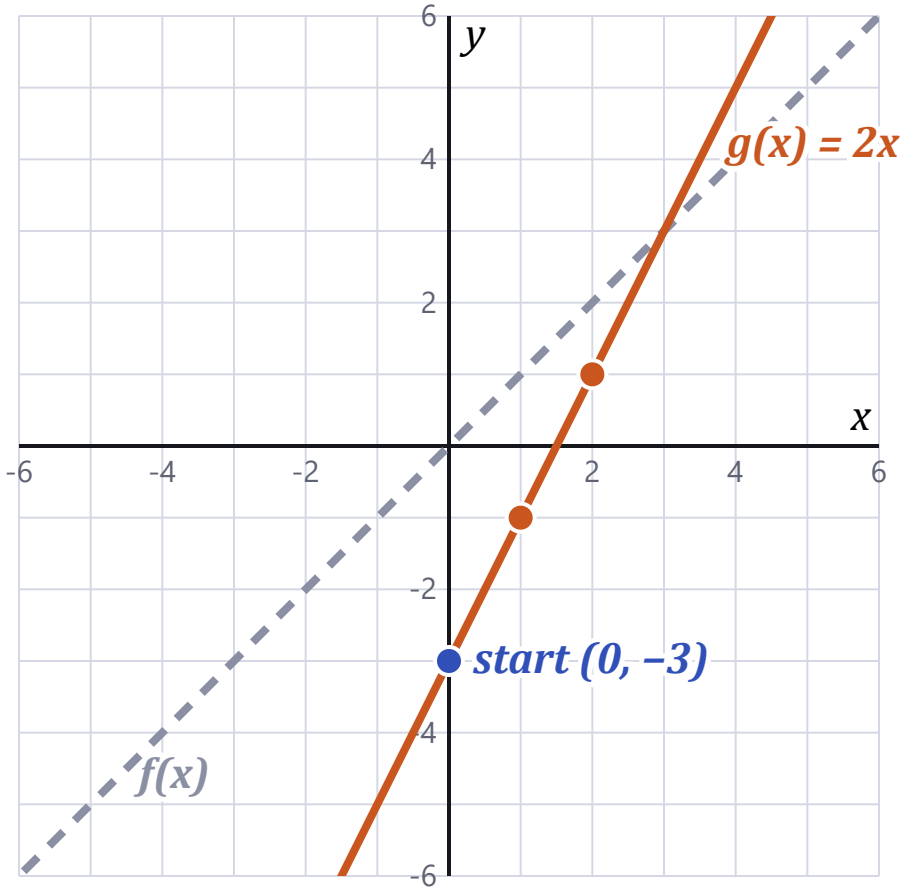
$$g(x) = a \cdot f(x) + d$$

a = slope

Stretch, compress or reflect

d = y-intercept

Up or down; line crosses at $(0, d)$



Example:
$g(x) = 2 \cdot f(x) - 3$

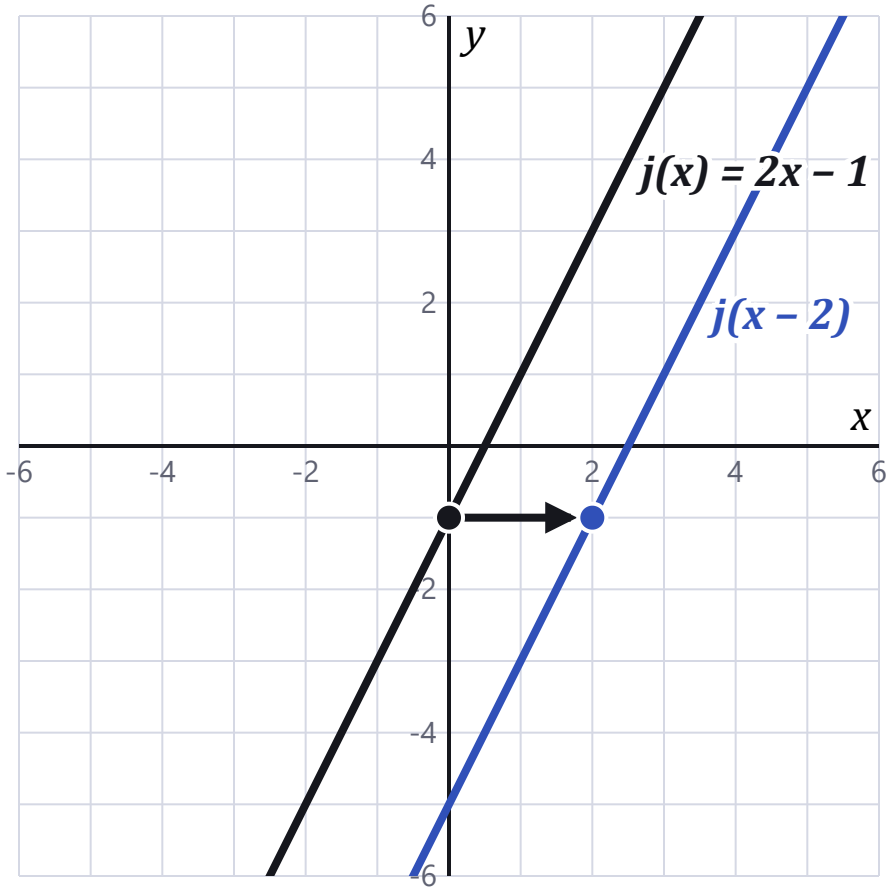
1. **Describe it:** stretch by 2, then down 3.
2. **Start** at the y-intercept $(0, -3)$.
3. **Use the slope** 2: up 2, right 1. Repeat.
4. **Draw** the line through the points.

Function	Describe it
$3f(x) - 1$	Stretch by 3, down 1
$-f(x) - 4$	Reflect across the x-axis, down 4

Horizontal Translation

$$y = f(x - c)$$

Inside the parentheses, the sign flips!



$f(x - 3)$

Moves **right** 3.
Minus inside → right.

$f(x + 3)$

Moves **left** 3.
Plus inside → left.

What changes

The **inputs** (x-values). Every point slides left or right.

What stays

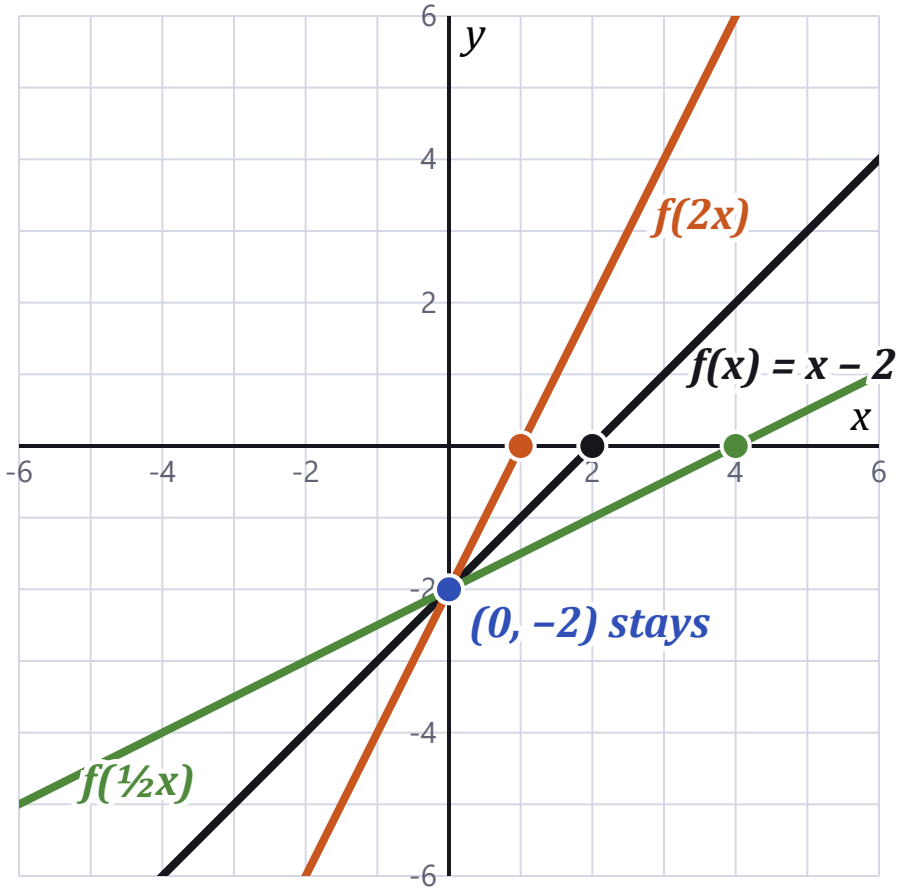
The slope. The new line is **parallel** to the original.

Example: $j(x) = 2x - 1$ right 2 →
$j(x - 2) = 2(x - 2) - 1 = 2x - 5$

Horizontal Dilation

$$y = f(bx)$$

Inside works backwards: a big *b* squeezes the graph.



$|b| > 1$

Compress: points move toward the *y*-axis.

$0 < |b| < 1$

Stretch: points move away from the *y*-axis.

$b < 0$

Reflect across the *y*-axis.

What changes

The **inputs** are multiplied by b.

The *x*-intercept moves: 2 → 1 → 4.

What stays

The *y*-intercept: points on the *y*-axis have $x = 0$.

Example: $f(x) = x - 2 \rightarrow f(2x) = 2x - 2$

Inside vs. Outside

$$y = a \cdot f(b(x - c)) + d$$

Outside: a and d

Change the **outputs** → **vertical**

- $+ d$: up d
- $- d$: down d
- a: vertical stretch or compress
- $a < 0$: reflect across the x-axis

Does what it says.

Inside: b and c

Change the **inputs** → **horizontal**

- $x - c$: right c
- $x + c$: left c
- b: horizontal compress or stretch
- $b < 0$: reflect across the y-axis

Does the opposite.

Only for $f(x) = x$!

Right n gives the same line as down n, and $n \cdot f(x)$ matches $f(nx)$.
For other lines they are **not** the same. $j(x) = 2x - 1$:

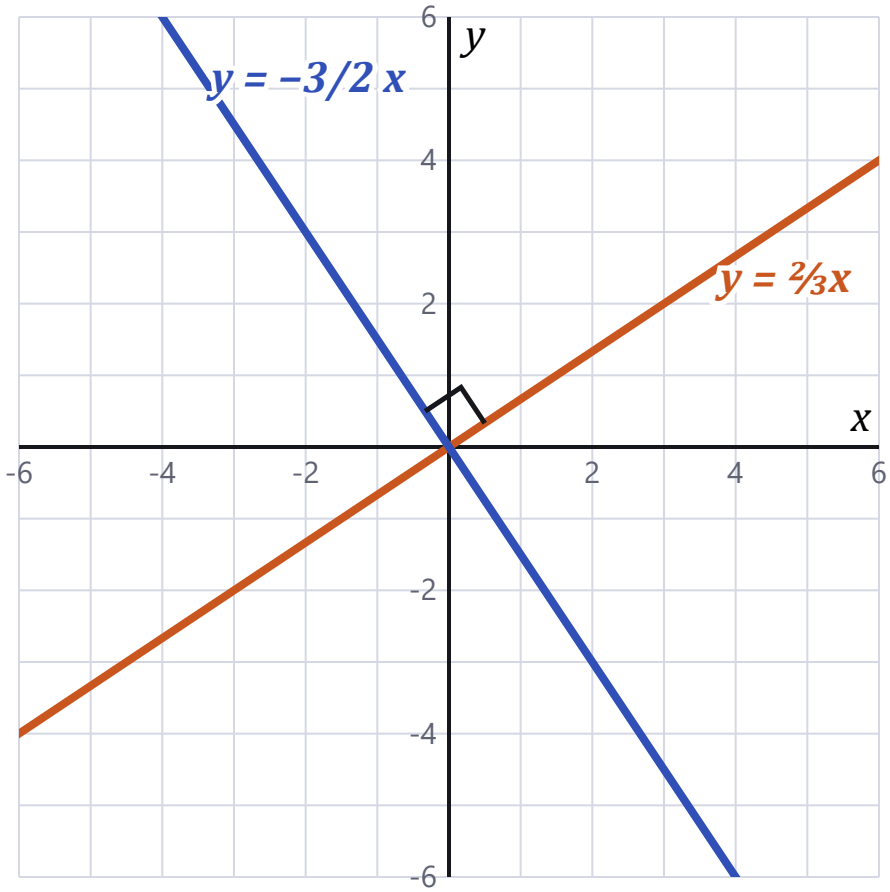
up 4: $2x + 3$ but left 4: $2x + 7$

> Example: $g(x) = -2 \cdot f(x - 3) + 1$
> right 3 · stretch by 2 · reflect across the x-axis · up 1

Perpendicular Lines

$$m_1 \cdot m_2 = -1$$

Flip the fraction. Change the sign.



Slope	⊥ slope
$\frac{2}{3}$	$-3/2$
2	$-\frac{1}{2}$
$-\frac{3}{4}$	$4/3$
-5	$\frac{1}{5}$
1	-1

Perpendicular ⊥

Meet at a **right angle**. Slopes are **negative reciprocals**: $\frac{2}{3} \cdot (-3/2) = -1$.

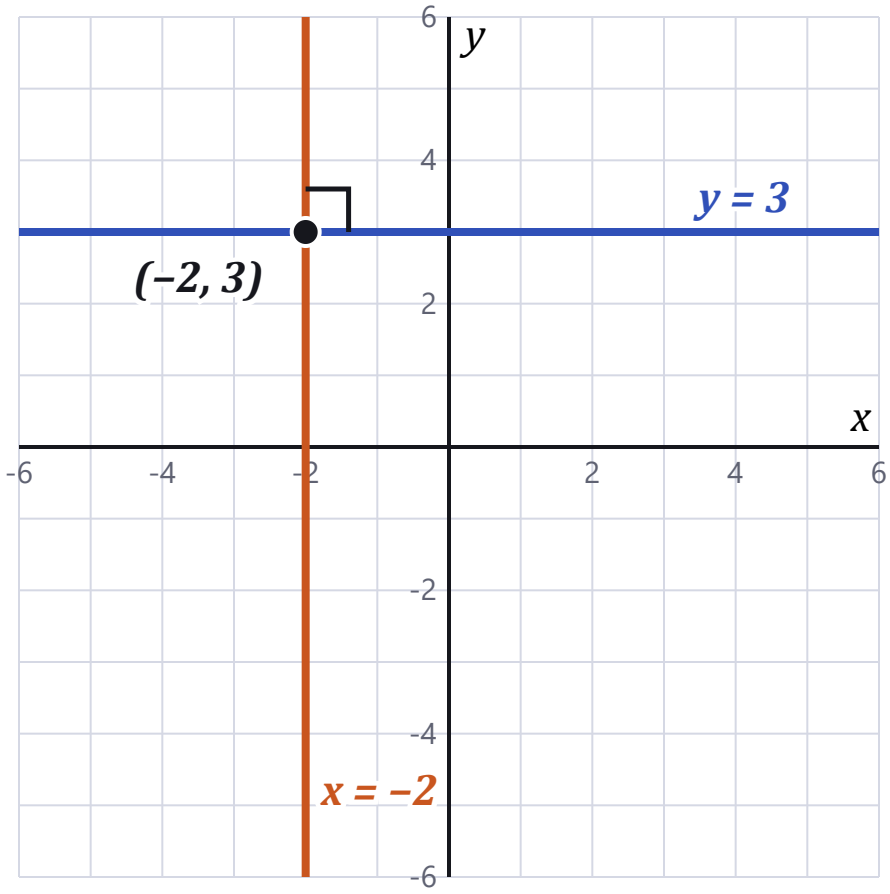
Parallel ∥

Never meet. Slopes are the **same**: $y = 2x + 1 \parallel y = 2x - 5$.

Why? A 90° turn maps (x, y) to $(-y, x)$, so slope b/a becomes $-a/b$.

Horizontal and Vertical Lines

HOY · VUX



H · O · Y

Horizontal

slope **0** (zero)

$y = k$

V · U · X

Vertical

slope **U**ndefined

$x = h$

Why?

Horizontal: y never changes, so slope $= 0 \div run = 0$.

Vertical: x never changes, so slope $= rise \div 0$: undefined.

Always

All horizontal lines are parallel. All vertical lines are parallel. A horizontal and a vertical line are **always perpendicular**.

Through $(2, -1)$: horizontal $y = -1$, vertical $x = 2$

Write a Parallel or Perpendicular Line

$$y - y_1 = m(x - x_1)$$

1. Find the **slope** of the given line.
2. **Parallel:** use the same slope. **Perpendicular:** flip it and change the sign.
3. **Plug in** the point: $y - y_1 = m(x - x_1)$ or $y = mx + b$.
4. **Solve** for y.

$\parallel$ to $y = -3x - 1$ through $(-1, 5)$

Same slope: $m = -3$

$5 = -3(-1) + b$

$5 = 3 + b$, so $b = 2$

$y = -3x + 2$

$\perp$ to $y = 2x + 1$ through $(6, 2)$

Flip and change: $m = -\frac{1}{2}$

$y - 2 = -\frac{1}{2}(x - 6)$

$y - 2 = -\frac{1}{2}x + 3$

$y = -\frac{1}{2}x + 5$

Check: put the point into your answer. It must make the equation true.